MILITARY SPECIFICATION SHEET

WAVEGUIDE ASSEMBLIES, RIGID, STRAIGHT, 90-DEGREE TWIST

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the waveguide assemblies described herein shall consist of this document and the latest issue of Specification MIL-W-3970.

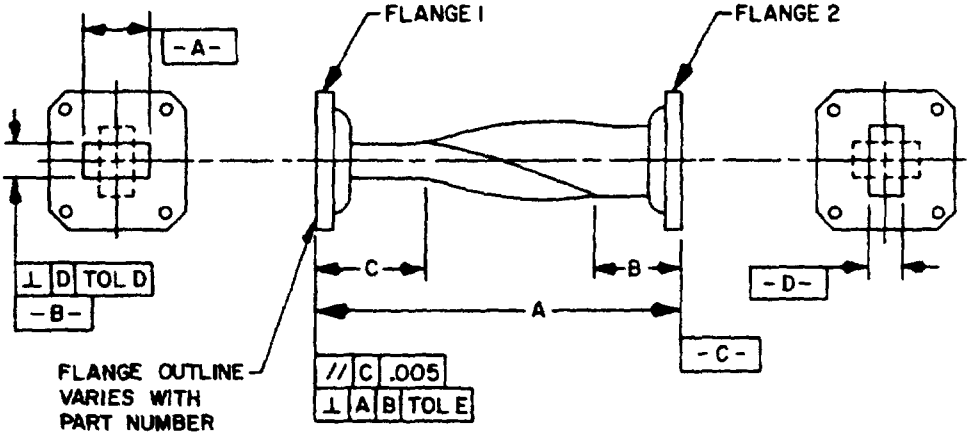


FIGURE 1 Assembly configuration

FSC 5985

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TABLE I Dash number, electrical and physical characteristics, and dimensions.

Dash No.	Waveguide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Pressurization lbf/in² gage	Dimensions 1/ 2/ 3/ A	B nom	C nom	D	E
				GHz	dB							
001	1-043	56-001	61-002	2.6 - 3.95	-	1.02	15	15.500 (393.70) ±.030 (.76)	1.375 (34.93)	.875 (22.23)	.025 (.64)	.012 (.30)
002	1-055	57-002	62-002	3.95 - 5.85	-	1.05	30	10.250 (260.35) ±.030 (.76)	1.000 (25.40)	.500 (12.70)	.016 (.41)	.008 (.20)
003	1-067	55-001	60-001	5.85 - 8.2	-	1.05	30	7.750 (196.85) ±.030	.812 (20.62)	.500 (12.70)	.012 (.30)	.006 (.15)
004	1-073	53-002	59-007	7.05 - 10	-	1.05	30	6.500 (165.10) ±.030 (.76)	.687 (17.45)	500 (12.70)	.010 (.25)	.005 (13)
005	1-079	53-001	59-006	8.2 - 12.4	-	1.05	30	5.250 (133.35) ±.025 (.64)	312 (7.92)	.500 (12.70)	.003 (08)	.004 (.10)
006	1-102	54-001	59-003	18 - 26.5	.11	1.06	30	2.563 (65.10) ±.015 (.38)	.312 (7.92)	.250 (6.35)	.004 (.10)	.002 (.05)
007	1-106	54-001	59-003	18 - 26.5	.11	1.06	30	2.563 (65.10) ±.015 (.38)	.312 (7.92)	.250 (6.35)	.004 (.10)	.002 (.05)
008	1-077	53-003	59-008	8.2 - 12.4	-	1.05	30	5.250 (133.35) ±.025 (.64)	.500 (12.70)	.500 (12.70)	.008 (.20)	.004 (.10)
009	1-071	53-004	59-009	7.05 - 10	-	1.05	30	6.500 (165.10) ±.030 (.76)	.687 (17.45)	.500 (12.70)	.010 (.25)	.005 (.13)
010	1-041	56-002	61-001	2.6 - 3.95	-	1.02	15	15.500 (393.70) ±.030 (.76)	1.375 (34.93)	.875 (22.23)	.012 (.30)	012 (.03)

See footnotes at end of table

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TABLE I. Dash number, electrical and physical characteristics, and dimensions - Continued

Dash No.	Waveguide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Pressurization lbf/in² gage	Dimensions 1/ 2/ 3/ A	B nom	C nom	D	E
				GHz	dB							
011	1-089	53-005	59-001	12 4 - 18	.07	1.05	30	3.125 (79.38) ±.020 (.51)	.437 (11 10)	.375 (9.53)	.006 (.15)	.003 (.08)
012	1-053	57-001	62-001	3.95 - 5 85	-	1.05	30	10.250 (260.35) ±.030 (.76)	1.000 (25.40)	.500 (12.70)	.016 (.41)	.008 (.20)
013	3-006	54-003	59-005	26 5 - 40	.15	1.10	30	1.813 (46.05) ±.015 (.38)	.250 (6.35)	.250 (6 35)	.002 (.05)	.001 (.03)
014	1-065	55-002	60-002	5.85 - 8.2	-	1.05	30	7.750 (196.85) ±.030 (.76)	.812 (20.62)	.500 (12.70)	.012 (.30)	006 (.15)
015	1-093	53-005	59-001	12.4 - 18	07	1.05	30	3.125 (79.38) ±.020 (.51)	.437 (11.10)	.375 (9.53)	.005 (.13)	.003 (.08)
016	1-103	54-002	59-004	18 - 26.5	11	1.06	30	2.563 (65.10) ±.015 (.38)	.312 (7.92)	.250 (6.35)	.004 (.10)	.002 (.05)
017	1-157	70-002	70-002	7.0 - 11	-	1.06	30	5.000 (127.00) ±.010 (.25)	.50 (12.70)	.50 (12.70)	.009 (.23)	.004 (.10)
018	1-073	59-007	59-007	7.05 - 10	-	1 05	30	6.000 (152.40) ±.010 (.25)	.687 (17.45)	.687 (17.45)	.010 (.25)	.005 (.13)
019	1-079	59-006	59-006	8.2 - 12.4	-	1.05	30	6.000 (152.40) ±.010 (.25)	.312 (7.92)	.312 (7.92)	.008 (.20)	.004 (.10)
020	1-090	53-006	59-002	12.4 - 18	.07	1.06	30	4.000 (101.60) ±.010 (.25)	.50 (12.70)	.50 (12.70)	.005 (.13)	.003 (.08)

See footnotes at end of table.

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TABLE I. Dash number, electrical and physical characteristics, and dimensions - Continued

Dash No.	Waveguide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Pressurization lbf/in^2 gage	Dimensions 1/ 2/ 3/ A	B nom	C nom	D	E
				GHz	dB							
021	1-090	53-006 4/	53-006 4/	12.4 - 18	.07	1.06	30	4.000 (101.60) ±.010 (.25)	.50 (12.70)	.50 (12.70)	.005 (.13)	.003 (.08)
022	3-006 3-008	54-003	54-003	26.5 - 40	.15	1.1	30	2.000 (50.80) ±.015 (.38)	.187 (4.75)	.285 (7.24)	.002 (.05)	.002 (.05)
023	3-006 3-008	67-005	67-005	26.5 - 40	.15	1.1	---	↓	.250 (6.35)	.250 (6.35)	.002 (.05)	.001 (.03)
024	3-010 3-012	67-006	67-006	33 - 50	.17		---	↓	↓	↓	.002 (.05)	.001 (.03)
025	3-014 3-016	67-007	67-007	40 - 60	.19	1.1	---	↓	↓	↓	.002 (.05)	.001 (.03)
026	3-017 3-019	67-008	67-008	50 - 75	.21	1.1	---	1.500 (38.10) ±.015 (.38)	↓	↓	.001 (.03)	.001 (.03)
027	3-020 3-022	67-009	67-009	60 - 90	.23	1.1	---	1.500 (38.10) ±.015 (.38)	↓	↓	.001 (.03)	.001 (.03)

1/ Dimensions are in inches.
2/ Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3/ Metric equivalents are in parentheses.
4/ Except flange thickness is .25 in lieu of .125.

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REQUIREMENTS

Dimensions and configuration: See figure 1 and table I.

Electrical characteristics: See table I.

Part number: M3970/1-(dash number from table I).

Cross reference: See table II.

TABLE II. Cross reference.

Part number M3970/1-	Type designation	Part number M3970/1-	Type designation
001	CG-992/U	009	CG-983/U
002	CG-994/U	010	CG-993/U
003	CG-980/U	011	CG-986/U
004	CG-982/U	012	CG-995/U
005	CG-984/U	013	CG-990/U
006	CG-988/U	014	CG-981/U
007	CG-989/U	015	CG-987/U
008	CG-985/U	016	CG-991/U

NOTE: Assemblies covered by this specification are substitutable for the following commercial types

Part number M3970/1-	FSN 5985-00-	Manufacturer's code	Commercial type
017		15869	723395-2
		17250	B-10878
018	529-0064	00443	590-1
019	425-3845	00443	690-1
020		05869	723395-1
020		17250	P9131
021		05869	712404-1
021		17250	712404-1
021		34441	62A-90T400-1
022		73293	44331H-900
023		73293	44332H-900
024		73293	44333H-900
025		73293	44334H-900
026		73293	44335H-900
027		73293	44336H-900

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Air Force - 85

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Navy - EC, OS
Air Force - 11, 17, 99
DSA - FS

User activities
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Navy - MC, CG, AS
Air Force - 14, 19

Preparing activity
Navy - EC

Agent:
DSA - ES

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